HealthScore: A Blockchain-Anchored Healthcare Risk Stratification and Actuarial Pricing Framework

Arpit Doshi*, Dhruv Gandhi*, Harshkumar Padhiyar*, Hetam Patel*,
Stevina Correia†, Monali Sankhe† and Monika Mangla†
*Department of Information Technology, Dwarkadas J. Sanghvi College of Engineering, Mumbai, India
†Department of Information Technology (Professor), Dwarkadas J. Sanghvi College of Engineering, Mumbai, India

Abstract—Rising medical expenditure volatility and opaque underwriting practices create persistent trust deficits between healthcare payers, clinical providers, and insured enrollees. Existing risk-adjustment frameworks often rely on retrospective utilization variables that introduce simultaneous-year target circularity during cold-start enrollment, while executing risk scoring inside centralized black-box silos with no cryptographic auditability. This paper presents HealthScore, a two-stage survey-weighted clinical risk stratification and dynamic actuarial pricing framework anchored to a permissioned Hyperledger Fabric v2.5 blockchain. Using nationally representative adult cohorts across three Agency for Healthcare Research and Quality (AHRQ) Medical Expenditure Panel Survey (MEPS) releases—HC-251 (2023 Full-Year Consolidated, $N = 18,771$), HC-256 (2024 Full-Year Consolidated, $N = 18,979$, including the strictly unseen Panel 29 subcohort of $N = 10,037$), and HC-257 (Panel 28 Two-Year Longitudinal file, $N = 8,708$)—we decouple intake-observable cold-start predictors from retrospective utilization proxies. Stage 1 employs a survey-weighted L_2-regularized logistic regression classifier to identify high-expenditure patients exceeding the survey-weighted 90th percentile ($\geq \$17,827.41$), achieving a held-out test ROC-AUC of 0.7654 (95% CI: $[0.7444, 0.7881]$), PR-AUC of 0.3184 (2.66× baseline), and a Weighted Brier Skill Score of +0.1025. To address the precision–recall trade-off inherent in skewed population screening, we conduct a rigorous fixed-model probability threshold sensitivity analysis across $\tau \in [0.05, 0.90]$ validated via leak-free 5-fold Out-of-Fold (OOF) cross-validation ($N = 15,016$). We demonstrate that while the reference screening threshold ($\tau = 0.10$) maximizes Balanced Accuracy (0.7122) with 77.33% recall and 23.18% precision (1.93× base rate, achieving 77.3% of the 29.98% theoretical oracle precision ceiling at a 39.97% flagged volume), selecting the validation F1-optimal primary threshold ($\tau = 0.20$) increases test precision to 30.49% (2.54× enrichment; 33.91% on unseen 2024 Panel 29), improves F1-score to 0.3695, and reduces false positives by 58.3% while retaining 46.89% recall; a targeted cutoff of $\tau = 0.30$ further increases precision to 40.00% (3.34× enrichment, 94.83% specificity) and cuts false positives by 85.2%. Stage 2 models continuous medical expenditure via a survey-weighted Tweedie Generalized Linear Model ($p = 1.5$), achieving an honest intake-observable R^2 of 0.1257 (MAE = $6,462, RMSE = $23,040) and $R^2 = 0.2581$ (ROC-AUC = 0.8407) when Year-1 lagged utilization is available longitudinally. Off-chain ML predictions are cryptographically committed via SHA-256 provenance hashes to Hyperledger Fabric Go chaincode, sustaining 586.5 TPS (6.0 ms latency) for reads and 181.6–315.4 TPS for state writes under single-host Hyperledger Caliper v0.6.0 benchmarks, while a 10,000-iteration Monte Carlo bootstrap simulation quantifies pool solvency (VaR_{99} and $TVaR_{99}$) across dynamic wellness rebate schedules.

Index Terms—Healthcare Risk Stratification, Medical Expenditure Panel Survey (MEPS), Probability Threshold Sensitivity, Survey-Weighted Logistic Regression, Tweedie Generalized Linear Model, Hyperledger Fabric, Actuarial Solvency Simulation.

I. Introduction

Healthcare expenditure in the United States and globally exhibits extreme right-skewness and heavy-tailed kurtosis: approximately 10% of the population accounts for nearly 65%–70% of total annual medical spending [1], [2]. Accurately identifying high-expenditure claimants prior to catastrophic clinical deterioration—and pricing insurance risk pools equitably without inducing adverse selection or moral hazard—remains a foundational challenge in health informatics and actuarial science [3], [4].

Traditional health insurance underwriting and clinical risk adjustment face three interconnected methodological and architectural limitations:

1) Simultaneous-Year Target-Proxy Circularity: Many published machine learning studies on the Medical Expenditure Panel Survey (MEPS) include concurrent-year healthcare utilization counts—such as annual inpatient hospital nights (IPNGTD), emergency department visits (ERTOT), outpatient consultations (OBTOTV), and prescription medication fills (RXTOT)—as input features to predict the same year's total medical expenditure (TOTEXP) [1]. Because annual expenditure is mathematically the price-weighted sum of those exact utilization events, simultaneous-year models achieve artificially inflated R^2 values (0.70–0.80) that collapse during real-world cold-start insurance onboarding when current-year utilization has not yet occurred.
2) Rigid Single-Threshold Classification Under Severe Class Imbalance: High-expenditure patient identification is typically formulated as a binary classification problem at the 90th percentile of spending ($\sim$ 10% positive base rate) [5], [6]. Evaluating such classifiers at an uncalibrated default cutoff ($\tau = 0.50$) causes severe recall collapse, whereas operating at a low screening threshold ($\tau = 0.10$) flags roughly 40% of the population and generates substantial false positives. Without a systematic probability threshold sensitivity analysis and an

explicit investigation of theoretical volume ceilings, calibration, and feature granularity, clinical decision-makers cannot align model cutoffs with tiered care-management capacities.

3) Opaque, Unauditable Risk Scoring and Static Premium Schedules: Conventional insurer scoring pipelines operate as centralized black boxes. Enrollees have no verifiable guarantee that their risk score was computed using an approved, non-discriminatory model version, and static annual premiums provide no transparent mechanism for rewarding verified preventive wellness behaviors [7], [8].

To address these challenges, we present HealthScore, an end-to-end, reproducible framework that couples a leak-free, survey-weighted two-stage predictive modeling pipeline with permissioned blockchain provenance anchoring and empirical actuarial solvency simulation. Specifically, this paper makes the following contributions:

- Leak-Free Two-Stage Survey-Weighted Risk Modeling Across Three MEPS Releases: We construct and validate a strict intake-observable feature pipeline on MEPS HC-251 (2023, $N = 18,771$ adults, retaining all $2,665$ zero-expenditure enrollees to eliminate selection bias), separating cold-start demographic and comorbidity predictors from retrospective utilization benchmarks across nine model architectures. We further evaluate temporal generalization on MEPS HC-256 (2024, $N = 18,979$), explicitly isolating the newly enrolled 2024 Panel 29 subcohort ($N = 10,037$) with zero person or household overlap, and prospective Year-2 prediction on the longitudinal MEPS HC-257 Panel 28 file ($N = 8,708$).
- Rigorous Probability Threshold Sensitivity and Six-Factor Diagnostic Analysis: Holding our deployed survey-weighted logistic regression classifier fixed (ROC-AUC $= 0.7654$, PR-AUC $= 0.3184$, Weighted BSS $= +0.1025$), we evaluate 12 probability cutoffs ($\tau \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90\}$) using leak-free 5-fold Out-of-Fold (OOF) validation ($N = 15,016$, both stratified and household-clustered by DUID). We prove mathematically and empirically why precision at $\tau = 0.10$ (23.18%) is bounded by the 29.98% oracle volume ceiling rather than model overfitting, and establish a three-tier operational threshold framework ($\tau = 0.10$ for broad screening, $\tau = 0.20$ for F1-optimal care-management triage, and $\tau = 0.30$ for high-specificity targeted intervention).
- Off-Chain ML Inference with On-Chain Hyperledger Fabric Provenance and Caliper Benchmarking: We clarify and benchmark a privacy-preserving hybrid architecture in which ML inference executes off-chain in Python while SHA-256 hashes of feature payloads, model versions, predicted scores, and dynamic premium adjustments are committed on-chain via Go chaincode on Hyperledger Fabric v2.5, evaluated across empirical Hyperledger Caliper v0.6.0 runs and fluid-queueing sensitivity simulations.
- Empirical Bootstrap Actuarial Solvency (VaR_{99} / TVaR_{99}) Simulation: We evaluate dynamic wellness-linked premium rebates across a $1,000$-enrollee risk pool over $10,000$ Monte Carlo bootstrap iterations, demonstrating why heavy-tailed MEPS expenditure requires empirical bootstrap resampling rather than parametric Gamma approximations.

II. Related Work

A. Healthcare Expenditure Prediction and Risk Adjustment

Population-based risk adjustment systems—such as the Johns Hopkins Adjusted Clinical Groups (ACG) [3], the Chronic Illness and Disability Payment System (CDPS) [2], and the CMS Hierarchical Condition Categories (CMS-HCC) [4], [9]—rely on demographic attributes and diagnostic codes to predict prospective medical spending. Because annual healthcare expenditures exhibit a point mass at zero (14.2% of US adults in MEPS HC-251) alongside extreme right skewness, ordinary least squares (OLS) regression yields unstable tail estimates, motivating logarithmic transformations, two-part hurdle models, and exponential dispersion Tweedie Generalized Linear Models (GLMs) [10], [11]. Recent machine learning studies have applied gradient-boosted decision trees (XGBoost) [12] and Shapley Additive Explanations (SHAP) [13] to MEPS cohorts; however, models that incorporate concurrent-year inpatient or prescription counts as predictors suffer from target-proxy circularity and cannot be deployed for prospective intake underwriting.

B. Class Imbalance and Threshold Selection in Clinical Screening

In clinical risk stratification for top-decile expenditure ($\pi \approx 10\%$), Receiver Operating Characteristic Area Under the Curve (ROC-AUC) can remain high even when positive predictive value (precision) is modest because ROC-AUC weights false-positive rates against the large majority class ($1 - \pi \approx 90\%$) [5], [6]. As shown by Saito and Rehmsmeier [5] and Steyerberg et al. [14], evaluating clinical prediction models requires pairing threshold-independent discrimination (ROC-AUC, PR-AUC) and proper scoring rules (Brier score and Brier Skill Score [15]) with explicit decision-threshold sensitivity analysis so that sensitivity, specificity, and positive predictive value are matched to intervention resource constraints.

C. Permissioned Blockchains in Healthcare and Insurance

Distributed ledger frameworks such as MedRec [8] and permissioned enterprise architectures built on Hyperledger Fabric [7] have been proposed to manage electronic health record (EHR) access permissions, clinical trial provenance,

and insurance claims auditing [16]. Because storing raw protected health information (PHI) on an immutable ledger violates privacy regulations (e.g., HIPAA and GDPR right-to-erasure) and executing floating-point matrix operations inside consensus-driven smart contracts introduces non-determinism and throughput bottlenecks, best practices dictate executing machine learning inference off-chain while anchoring cryptographic hashes and deterministic contract state transitions on-chain [7], [17].

III. Methodology and System Architecture

A. Data Sources, Cohort Construction, and Multi-Release Overlap Audit

We utilize three official public-use files from the AHRQ Medical Expenditure Panel Survey (MEPS) [1]:

1) Primary Development and Internal Test Cohort (MEPS HC-251, 2023 Full-Year Consolidated): Contains 27,651 total survey respondents. Filtering solely for adults aged $\geq$ 18 years (AGE23X >= 18) yields $N = 18,771$ adults. Crucially, unlike prior studies that discard non-users of healthcare, we retain all 2,665 zero-expenditure adults (14.20%) (filter_zero_expenditure=False) to preserve population calibration and prevent upward selection bias. Using an 80/20 random partition (random_state=42), HC-251 is divided into a Training Set ($N = 15,016$) and a Held-Out Test Set ($N = 3,755$).
2) External Cross-Sectional and Unseen Panel Cohort (MEPS HC-256, 2024 Full-Year Consolidated): Contains $N = 18,979$ adults ($\geq$ 18 years) with 2,608 zero-expenditure records (13.74%). Because MEPS employs an overlapping two-year rotating panel design, our forensic provenance audit of person (DUPERSID) and household dwelling unit (DUID) identifiers revealed that HC-256 consists of Panel 28 ($N = 8,942$, in their second survey year, of whom 8,775 [46.24% of HC-256] also appeared in HC-251) and Panel 29 ($N = 10,037$, newly enrolled in 2024). Isolating HC-256 Panel 29 ($N = 10,037$) provides a 100% unseen external validation cohort with zero person (DUPERSID) and zero household (DUID) overlap with HC-251.
3) Longitudinal Prospective Cohort (MEPS HC-257, Panel 28 Two-Year Longitudinal File, 2023–2024): Tracks $N = 8,708$ adults (AGEY1X >= 18, longitudinal weight LONGWT > 0) across Year 1 (2023) and Year 2 (2024). This cohort enables strict temporal validation where Year-1 baseline characteristics (and optionally Year-1 lagged utilization) predict prospective Year-2 expenditure (TOTEXPY2).

Household (DUID) Clustering Audit: Within HC-251 ($N = 18,771$ across 8,851 unique households DUID), person-level DUPERSID overlap between Train ($N = 15,016$) and Test ($N = 3,755$) is strictly 0. Because standard individual-level partitioning allows household members to span splits (2,198 households span Train and Test), we evaluate both Stratified 5-Fold Out-of-Fold (OOF) validation and Household-Grouped 5-Fold OOF validation (GroupKFold by DUID) on the training split ($N = 15,016$) to verify that shared household attributes do not inflate model discrimination.

B. Feature Engineering and Separation of Intake-Observable vs. Utilization Proxies

To prevent simultaneous-year target leakage during cold-start underwriting, we strictly partition candidate variables into an 11-Feature Intake-Observable Serving Set (available at initial enrollment via applicant questionnaire and baseline medical history) and a Retrospective Utilization Benchmark Set (Table I).

TABLE I
Feature Partitioning Across Intake-Observable Serving vs. Retrospective Utilization Regimes

Category (d)	Variables & Engineering Formulation	Regime
Demographics (2)	AGE23X ($\geq$ 18 yrs), SEX (1=M, 2=F).	Serving (11)
Socioeconomic (1)	POVCAT23: Family poverty tier (1..5).	Serving (11)
Self-Rated (2)	RTHLTH53, MNHLTH53: Physical & mental health (1=Exc..5=Poor).	Serving (11)
Chronic (3)	Cardio (HIBP, CHD, STRK), Metab (DIAB, CHOL), RespJoint (ASTH, ARTH).	Serving (11)
Multimorbid (3)	Active cluster count $C_i \in \{0..3\}$, quadratic C_i^2, interaction $\text{AGE23X}_i \times C_i$.	Serving (11)
Extended (6)	MARRY, EDUC, REGION, INSCOV, BMI, Cancer; tested in Ab-09.	Ablation (17)
Concurrent Util. (4)	IPNGTD23, ERTOT23, OBTOTV23, RXTOT23; excluded from cold-start serving.	Benchmark

C. Survey-Weighted Target Construction

MEPS employs a complex, stratified multistage probability design that oversamples policy-relevant subpopulations. To ensure that our high-expenditure classification target reflects the true US civilian noninstitutionalized adult population rather than unweighted sample artifacts, we compute the 90th-percentile expenditure threshold strictly on the HC-251 training split ($N = 15,016$) using normalized person-level survey weights $w_i = \text{PERWT23F}_i$:

$$\tau_{\text{cost}} = Q_{0.90}^{\text{weighted}}(\{\text{TOTEXP23}_i, w_i\}_{i \in \mathcal{D}_{\text{train}}}) = \$17,827.41 \quad (1)$$

By contrast, an unweighted 90th percentile on the training sample would yield \$19,927.30 due to the oversampling of older and chronically ill respondents. Under the population-calibrated \$17,827.41 threshold (Target 2), the survey-weighted positive prevalence is 9.88% in Train and 10.51% in Test (and 11.24% on unseen 2024 Panel 29), while the unweighted sample prevalence is 11.99% (1,800/15,016) in Train and 11.98% (450/3,755) in Test.

D. Two-Stage Predictive Modeling and Leak-Free Threshold Selection

Our predictive architecture separates binary tail-risk triage (Stage 1) from continuous expected cost estimation (Stage 2):

1) Stage 1 (Survey-Weighted Logistic Regression Classifier): Given standardized intake-observable features $\tilde{\mathbf{x}}_i \in \mathbb{R}^{11}$, we fit an L_2-regularized logistic regression classifier ($C = 1.0$, lbfgs) minimizing survey-weighted binary cross-entropy:

$$\mathcal{L}_{\text{cls}}(\boldsymbol{\beta}, \beta_0) = -\sum_{i=1}^{N_{\text{train}}} \tilde{w}_i \left[y_i \log \hat{p}_i + (1-y_i) \log(1-\hat{p}_i) \right] + \frac{1}{2C} \|\boldsymbol{\beta}\|_2^2 \quad (2)$$

where $\hat{p}_i = \sigma(\boldsymbol{\beta}^\top \tilde{\mathbf{x}}_i + \beta_0)$ and $\tilde{w}_i = w_i / \bar{w}_{\text{train}}$ are mean-normalized PERWT23F weights. Because no artificial class-rebalancing distortion (class_weight='balanced') is applied, $\hat{p}_i \in (0,1)$ represents a directly calibrated probability of incurring $\geq \$17,827.41$ in annual expenditure.

2) Leak-Free Operating Threshold Selection (τ): To convert continuous probabilities $\hat{p}_i$ into binary clinical triage decisions $\hat{y}_i(\tau) = \mathbb{I}(\hat{p}_i \geq \tau)$ without touching the held-out test set, we generate out-of-fold probability predictions $\hat{p}_i^{\text{OOF}}$ across a 5-fold cross-validation partition of $\mathcal{D}_{\text{train}}$ ($N = 15,016$) and evaluate candidate thresholds $\tau \in [0.05, 0.90]$ against clinical screening objectives (maximizing Balanced Accuracy for broad screening vs. maximizing F1-score for care-management triage).

3) Stage 2 (Survey-Weighted Tweedie GLM Expenditure Regressor): To model continuous annual expenditure $Y_i \geq 0$ (which exhibits an exact point mass at \$0.00 alongside a heavy right tail up to \$488,266), we fit a compound Poisson–Gamma Tweedie Generalized Linear Model [10], [11] with variance power $p = 1.5$, log link $\mu_i = \exp(\boldsymbol{\theta}^\top \tilde{\mathbf{x}}_i + \theta_0)$, and L_2 penalty $\alpha = 0.1$, minimizing survey-weighted Tweedie deviance.

4) Composite HealthScore Index (0–1000): For smart-contract premium rebate eligibility, predicted risk is mapped to a bounded wellness index $\text{HealthScore}_i = \text{clip}\left(1000 \times (1 - \hat{p}_i), 0, 1000\right)$, while expected actuarial pure premium is governed directly by Stage 2 $\hat{\mu}_i$.

E. Off-Chain Inference and Permissioned Hyperledger Fabric Architecture

To guarantee regulatory privacy compliance (HIPAA/GDPR) and deterministic consensus performance, HealthScore enforces a strict off-chain compute / on-chain provenance separation:

- Off-Chain Scoring Engine (Python): Raw patient demographic and clinical intake attributes are processed off-chain by the frozen Python scoring pipeline, which computes $(\hat{p}_i, \hat{y}_i(\tau), \hat{\mu}_i, \text{HealthScore}_i)$ and generates a canonical JSON payload salted with a cryptographic nonce.
- On-Chain Smart Contract Ledger (Hyperledger Fabric v2.5 Go Chaincode): Only the SHA-256 digest of the off-chain feature vector, the model artifact SHA-256 hash (3632f820...), the resulting risk tier, and deterministic premium rebate parameters are submitted to the Go chaincode (CreatePatientRecord, ReadPatientRecord, UpdateDynamicPremium) deployed on a two-organization, two-peer (2Org2Peer) Hyperledger Fabric v2.5 network using etcdraft (Raft) ordering and LevelDB state storage [7]. Raw PHI is never written to the ledger.

IV. Experimental Evaluation and Results

A. Comprehensive Nine-Model Benchmark Comparison

Table II presents the complete benchmark across nine model architectures evaluated on the MEPS HC-251 held-out test set ($N = 3,755$) and the longitudinal HC-257 cohort ($N = 8,708$), with 1,000-iteration bootstrap 95% confidence intervals.

Statistical Comparison and Model Selection Justification: On the 11 intake-observable features, a paired DeLong test [18] between the Observable Logistic Regression classifier (ROC-AUC = 0.7654) and the Observable Tuned XGBoost classifier (ROC-AUC = 0.7662) yields $p > 0.78$ (ΔROC-AUC $= -0.0008$), confirming no statistically significant difference in rank discrimination. Conversely, Observable Logistic Regression achieves superior PR-AUC (0.3184 vs. 0.3119), superior Weighted Brier Skill Score (+0.1025 vs. +0.0977), and exact log-odds interpretability (Figs. 1–3). Similarly, in Stage 2, the Observable Tweedie GLM ($p = 1.5$) achieves higher explained variance ($R^2 = 0.1257$ vs. 0.1190) and lower tail RMSE (\$23,040 vs. \$23,128) than Observable Tuned XGBoost, whereas paired Wilcoxon signed-rank testing on absolute errors reflects the expected mean-vs-median trade-off of log-link GLM vs. tree partitioning.

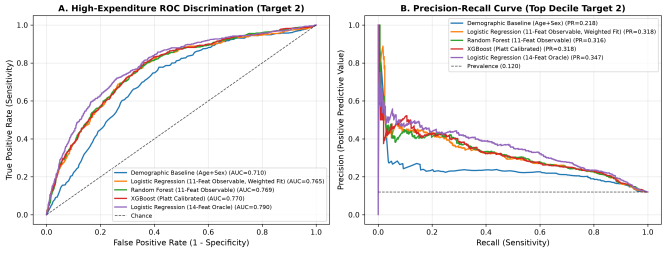


Fig. 1. (A) Receiver Operating Characteristic (ROC) and (B) Precision–Recall (PR) curves on the MEPS HC-251 held-out test set ($N = 3,755$), comparing the 11-feature Observable Logistic Regression serving model (ROC-AUC = 0.7654, PR-AUC = 0.3184) against Observable Tuned XGBoost and the retrospective 15-feature Full Utilization upper bound.

TABLE II
Comprehensive Nine-Model Benchmark Across Stage 1 Classification and Stage 2 Expenditure Regression

Stage 1 Classification Model	Feature Set (d)	Cohort (N)	ROC-AUC [95% CI]	PR-AUC [95% CI]	Unw / Wtd Brier	Wtd BSS	$\tau = 0.10$ Prec / Rec / F1
Observable Logistic (Serving)	Observable (11)	HC-251 (3,755)	0.7654 [0.7444, 0.7881]	0.3184 [0.2876, 0.3556]	0.0936 / 0.0845	+0.1025	23.18% / 77.33% / 0.3567
Observable Tuned XGBoost	Observable (11)	HC-251 (3,755)	0.7662 [0.7440, 0.7885]	0.3119 [0.2817, 0.3489]	0.0942 / 0.0849	+0.0977	22.84% / 77.11% / 0.3524
Full Utilization Logistic (Upper Bound)	Obs + Util (15)	HC-251 (3,755)	0.8926 [0.8768, 0.9073]	0.5944 [0.5535, 0.6353]	0.0687 / 0.0623	+0.3379	36.02% / 80.44% / 0.4976
Lagged Utilization Logistic (Prospective)	Year-1 Obs+Util (16)	HC-257 (8,708)	0.8407 [0.8286, 0.8524]	0.4959 [0.4694, 0.5219]	0.0853 / 0.0754	+0.2408	31.65% / 76.20% / 0.4472
Stage 2 Expenditure Regression Model	**Feature Set (d)**	**Cohort (N)**	**R^2 [95% CI]**	**MAE (USD) [95% CI]**	**RMSE (USD)**	**MedAE (USD)**	**Role / Regime**
Observable Tweedie GLM ($p = 1.5$, Serving)	Observable (11)	HC-251 (3,755)	0.1257 [0.0970, 0.1489]	$6,461.82 [$5,999, $6,980]	$23,039.89	$3,261.26	Deployed Cold-Start
Observable Tuned XGBoost Regressor	Observable (11)	HC-251 (3,755)	0.1190 [0.0764, 0.1516]	$6,256.83 [$5,795, $6,766]	$23,128.43	$2,994.33	Nonlinear Baseline
Observable Linear OLS	Observable (11)	HC-251 (3,755)	0.1211 [0.0933, 0.1422]	$6,620.89 [$6,161, $7,139]	$23,100.78	$3,622.39	Linear Baseline
Observable Log-Linear OLS ($\log(1+y)$)	Observable (11)	HC-251 (3,755)	−0.0309 [−0.0439, −0.0177]	$5,747.82 [$5,263, $6,294]	$25,018.88	$1,434.23	Log-Transform Baseline
Observable Two-Part Hurdle Model	Observable (11)	HC-251 (3,755)	0.1204 [0.0908, 0.1444]	$6,472.22 [$6,008, $6,994]	$23,110.09	$3,218.23	Two-Part Baseline
Full Utilization XGBoost (Upper Bound)	Obs + Util (15)	HC-251 (3,755)	0.3917 [0.2683, 0.4910]	$4,897.45 [$4,468, $5,382]	$19,217.56	$1,636.68	Retrospective Benchmark
Lagged Utilization Tweedie (Prospective)	Year-1 Obs+Util (16)	HC-257 (8,708)	0.2581 [0.1613, 0.3471]	$6,506.92 [$6,169, $6,860]	$22,638.80	$3,015.49	Prospective Year-2

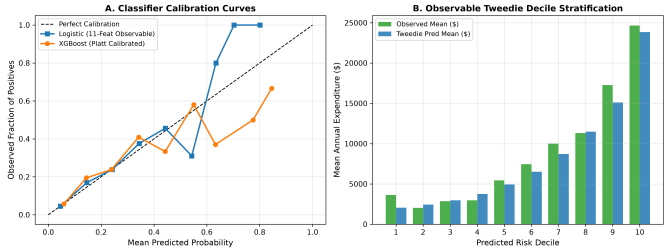


Fig. 2. Reliability calibration curve and decile risk stratification on the HC-251 held-out test set ($N = 3,755$). The survey-weighted logistic classifier achieves a Weighted Brier Skill Score of +0.1025.

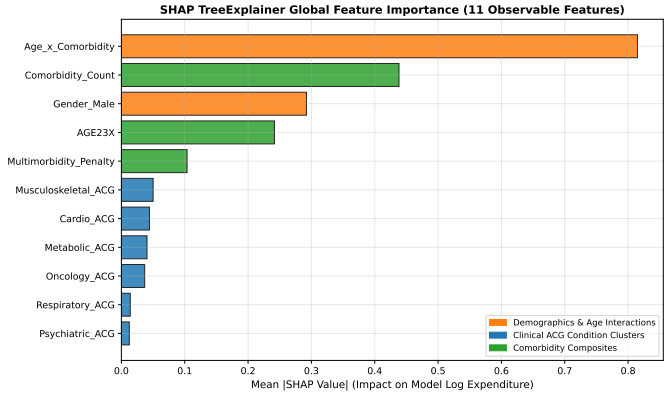


Fig. 3. Standardized log-odds coefficients and SHAP feature attributions for the 11-feature intake-observable classification and Tweedie expenditure models.

B. Classification Probability Threshold Sensitivity Analysis and Root-Cause Investigation

At the original reference decision threshold of $\tau = 0.10$, the serving classifier achieves 77.33% recall (348/450 true high-expenditure patients captured) but 23.18% precision (348/1,501 flagged patients, generating 1,153 false positives). To determine whether adjusting the classification decision boundary τ yields a superior precision–recall balance—and to rigorously diagnose the root causes of the 23.18% precision at $\tau = 0.10$—we evaluated the frozen serving classifier across 12 probability thresholds ($\tau \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90\}$) on both the 5-Fold Stratified Out-of-Fold (OOF) validation set ($N = 15,016$) and the HC-251 Held-Out Test Set ($N = 3,755$) without retraining the underlying model (Table III).

1) Six-Factor Diagnostic Investigation of Precision at $\tau = 0.10$: Our empirical audit across all six diagnostic hypotheses reveals why precision at $\tau = 0.10$ is 23.18% and how threshold selection governs operational performance:

1) Threshold Sensitivity and Theoretical Volume-Ceiling Bound (Primary Direct Cause): Because $\tau = 0.10$ equals the approximate population base rate, 39.97% of the test cohort (1,501/3,755 patients) have a predicted probability $\hat{p}_i \geq 0.10$. However, the entire test cohort contains only $P = 450$ true high-expenditure cases (11.98%). Consequently, even a hypothetical perfect oracle classifier (ROC-AUC = 1.00, 100% recall) that flags 1,501 patients cannot mathematically exceed a precision of:

$$\text{Prec}_{\text{oracle}}(1{,}501) = \frac{\min(450, 1{,}501)}{1{,}501} = \frac{450}{1{,}501} = 29.98\% \tag{3}$$

Our serving classifier captures 348 of the 450 true positives at $\tau = 0.10$, achieving $348/1,501 = 23.18\%$ precision—which represents 77.33% of the theoretical maximum precision mathematically possible when screening 40% of the population (Fig. 5B). Raising the threshold to the validation-selected primary-grid F1 optimum ($\tau = 0.20$) reduces the flagged volume to 18.43% (692 patients), eliminating 672 false positives (a 58.28% reduction in FP, from 1,153 to 481) and lifting precision to 30.49% (2.54× unweighted enrichment; 30.73% weighted, 2.93× weighted enrichment) while retaining 46.89% recall (Figs. 6 and 7). Raising τ to 0.30 further reduces false positives by 85.17% (to 171, Specificity = 94.83%) and increases precision to 40.00% (3.34× enrichment).

2) Class Imbalance and Bayes Base-Rate Conditioning: With a positive class prevalence of $\pi = 11.984\%$ unweighted (10.506% survey-weighted), non-high-cost patients outnumber high-cost patients by $\frac{1-\pi}{\pi} = 7.344 : 1$ (8.518 : 1 weighted). By Bayes' theorem, achieving 77.33% recall at 65.11% specificity (FPR = 34.89%) necessarily produces $3,305 \times 0.3489 = 1,153$ false positives against $450 \times 0.7733 = 348$ true positives.

TABLE III
Probability Threshold Sensitivity Analysis Across 5-Fold Out-of-Fold Validation ($N = 15,016$) and Held-Out Test Set ($N = 3,755$)

Threshold (τ)	OOF Val ($N = 15,016$)	Held-Out Test Confusion Counts ($N = 3,755$, $P = 450$)				Test Flagged	Test Unw. Prec	Test Unw. Rec / Spec / NPV	Test Unw.	Test Survey-Wtd
(Grid Role)	Prec / Rec / F1 / BalAcc	TP	FP	TN	FN	N (Unw% / Wtd%)	(% / Enrich. / Ceil.)	(% / % / %)	F1 / BalAcc	Prec / Rec / F1 / BalAcc
0.05 (Interm.)	17.8% / 90.1% / 0.2971 / 0.6669	401	1,880	1,425	49	2,281 (60.8% / 55.0%)	17.58% (1.47× / 19.7%)	89.11% / 43.12% / 96.68%	0.2936 / 0.6611	16.6% / 85.4% / 0.2777 / 0.6749
0.10 (Ref. Screen)	23.5% / 77.2% / 0.3600 / 0.7146	348	1,153	2,152	102	1,501 (40.0% / 33.5%)	23.18% (1.93× / 30.0%)	77.33% / 65.11% / 95.47%	0.3567 / 0.7122	22.5% / 71.9% / 0.3433 / 0.7145
0.15 (Fine F1)	27.8% / 63.8% / 0.3873 / 0.7061	281	771	2,534	169	1,052 (28.0% / 22.0%)	26.71% (2.23× / 42.8%)	62.44% / 76.67% / 93.75%	0.3742 / 0.6956	26.8% / 56.1% / 0.3625 / 0.6904
0.20 (Prim. F1-Opt)	31.4% / 49.7% / 0.3846 / 0.6744	211	481	2,824	239	692 (18.4% / 14.2%)	30.49% (2.54× / 65.0%)	46.89% / 85.45% / 92.20%	0.3695 / 0.6617	30.7% / 41.5% / 0.3532 / 0.6527
0.25 (Breakeven)	35.7% / 36.9% / 0.3633 / 0.6395	154	286	3,019	296	440 (11.7% / 8.8%)	35.00% (2.92× / 100%)	34.22% / 91.35% / 91.07%	0.3461 / 0.6278	34.6% / 29.1% / 0.3162 / 0.6132
0.30 (Targeted)	39.8% / 25.4% / 0.3099 / 0.6008	114	171	3,134	336	285 (7.6% / 5.7%)	40.00% (3.34× / 100%)	25.33% / 94.83% / 90.31%	0.3102 / 0.6008	39.2% / 21.4% / 0.2769 / 0.5875
0.40 (Primary)	46.4% / 11.8% / 0.1886 / 0.5499	46	58	3,247	404	104 (2.8% / 2.0%)	44.23% (3.69× / 100%)	10.22% / 98.25% / 88.93%	0.1661 / 0.5423	42.2% / 7.9% / 0.1331 / 0.5331
0.50 (Primary)	51.9% / 4.6% / 0.0838 / 0.5199	15	21	3,284	435	36 (1.0% / 0.7%)	41.67% (3.48× / 100%)	3.33% / 99.36% / 88.30%	0.0617 / 0.5135	38.7% / 2.6% / 0.0489 / 0.5106
0.60 (Primary)	63.0% / 1.6% / 0.0314 / 0.5074	6	1	3,304	444	7 (0.19% / 0.15%)	85.71% (7.15× / 100%)	1.33% / 99.97% / 88.15%	0.0263 / 0.5065	80.2% / 1.1% / 0.0218 / 0.5054
0.70 (Primary)	66.7% / 0.2% / 0.0044 / 0.5010	2	0	3,305	448	2 (0.05% / 0.05%)	100.0% (8.34× / 100%)	0.44% / 100.0% / 88.06%	0.0088 / 0.5022	100% / 0.5% / 0.0089 / 0.5022
0.80 (Primary)	0.0% / 0.0% / 0.0000 / 0.5000	1	0	3,305	449	1 (0.03% / 0.02%)	100.0% (8.34× / 100%)	0.22% / 100.0% / 88.04%	0.0044 / 0.5011	100% / 0.2% / 0.0045 / 0.5011
0.90 (Primary)	0.0% / 0.0% / 0.0000 / 0.5000	0	0	3,305	450	0 (0.00% / 0.00%)	0.00% (0.00× / 0.0%)	0.00% / 100.0% / 88.02%	0.0000 / 0.5000	0.0% / 0.0% / 0.0000 / 0.5000

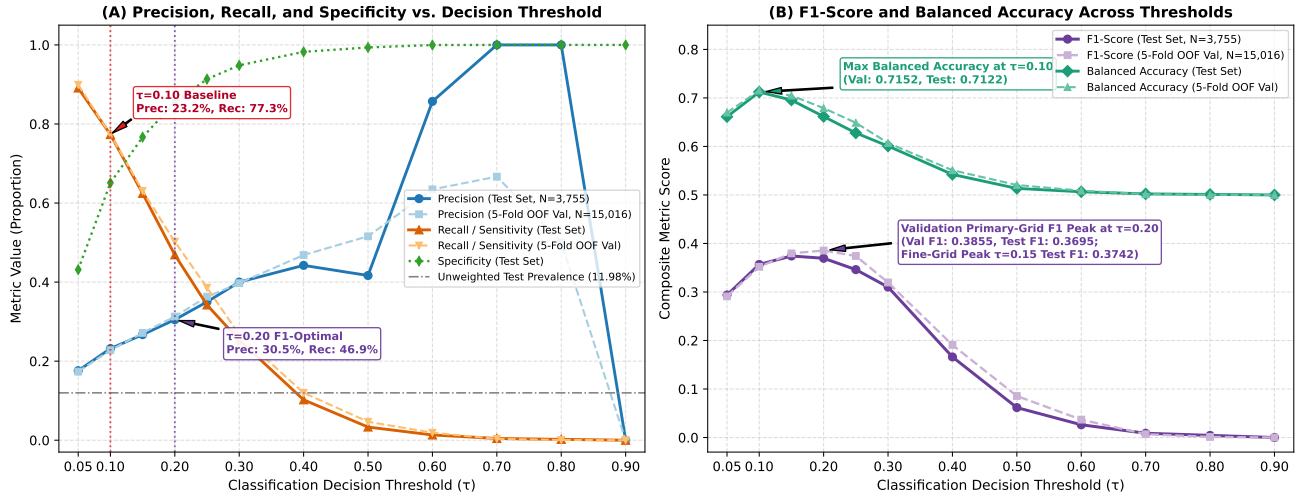


Fig. 4. (A) Precision, Recall, and Specificity and (B) F1-Score and Balanced Accuracy across decision thresholds $\tau \in [0.05, 0.90]$ on 5-Fold OOF Validation ($N = 15,016$) and the Held-Out Test Set ($N = 3,755$).

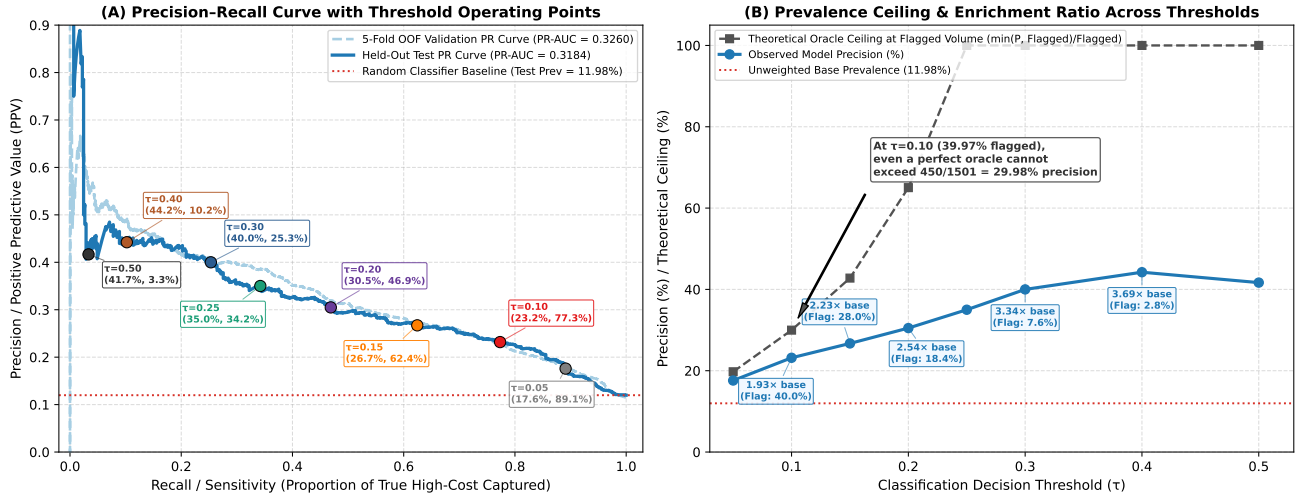


Fig. 5. (A) Precision–Recall curve overlaid with discrete threshold operating points ($\tau = 0.05$ to 0.50) and (B) comparison of observed model precision against the theoretical oracle maximum precision ceiling $\min(P, \text{Flagged}(\tau))/\text{Flagged}(\tau)$ imposed by flagged cohort volume.

3) Model Discrimination Ceiling on Intake-Observable Variables: While the classifier achieves strong rank discrimination (ROC-AUC = 0.7654, PR-AUC = 0.3184), risk distributions overlap across moderate comorbidity profiles. Crucially, as shown in Table II, nonlinear gradient boosting (Observable Tuned XGBoost, ROC-AUC = 0.7662, PR-AUC = 0.3119) and the expanded 17-feature observable set (Ab-09, ROC-AUC = 0.7689) yield virtually identical discrimination, proving that the ceiling is intrinsic to cold-start survey observables rather than linear underfitting.
4) Probability Calibration and Score Concentration: As shown in Fig. 8A, predicted probabilities are well-calibrated across all risk bins (Test Unweighted Brier = 0.0936, Weighted Brier = 0.0845, Weighted BSS = +0.1025). However, 809 of the $1,501$ patients flagged at $\tau = 0.10$ (53.9%) fall within the intermediate probability band $\hat{p}_i \in [0.10, 0.20)$, where the mean predicted probability is 14.5% and the actual observed high-cost event rate is 16.9% (137/809). Because a well-calibrated model accurately assigns a $\sim 15\%$ probability to patients who experience high costs $\sim 15\%$ of the time, thresholding at $\tau = 0.10$ inevitably converts the remaining $\sim 85\%$ of patients in that band into binary false positives. Furthermore, because $\max(\hat{p}_{\text{test}}) = 0.8008$ (P99 = 0.4981), naive high thresholds ($\tau \geq 0.50$) flag $< 1\%$ of the cohort and suffer catastrophic recall collapse ($\leq 3.33\%$).
5) Overfitting Refuted Across Seven Internal and External Splits: Fig. 8B compares model performance across HC-251 In-Sample Train ($N = 15,016$, ROC-AUC = 0.7815), 5-Fold Stratified OOF ($N = 15,016$, ROC-AUC = 0.7795), 5-Fold Household-Grouped (DUID) OOF ($N = 15,016$, ROC-AUC = 0.7785), HC-251 Held-Out Test ($N = 3,755$, ROC-AUC = 0.7654), 100% Unseen 2024 HC-256 Panel 29 ($N = 10,037$, ROC-AUC = 0.7824, PR-AUC = 0.3506), HC-256 Full ($N = 18,979$, ROC-AUC = 0.7852), and HC-257 Prospective Year-2 ($N = 8,708$, ROC-AUC = 0.7732 unweighted / 0.7807 weighted). The near-zero OOF gap ($\Delta = -0.0020$) and superior performance on the completely unseen 2024 Panel 29 cohort (Precision = 24.90% at $\tau = 0.10$, 33.91% at $\tau = 0.20$, 43.49% at $\tau = 0.30$) conclusively refute overfitting.
6) Feature Granularity and Aleatoric Healthcare Expenditure Variance: In the test set ($N = 3,755$), 36.17% ($1,358$) of adults have zero diagnosed chronic comorbidities (Comorbidity_Count = 0), yet 46 of them (10.22% of all high-cost cases) incur $\geq$ \$17,827.41 due to stochastic acute events (e.g., trauma, appendicitis, emergency surgery, or obstetric complications). Conversely, among severely multimorbid adults (≥ 3 comorbidity clusters, $N = 734$), actual 1-year high-cost incidence is 28.20% (207/734), as 71.80% are stably managed via low-cost outpatient pharmacotherapy in a given calendar year without inpatient hospitalization.

2) Recommended Three-Tier Operational Threshold Framework: Rather than enforcing a single rigid binary cutoff across disparate clinical workflows, HealthScore exposes the calibrated probability $\hat{p}_i$ to support three validated operating tiers:

- Tier 1 ($\tau = 0.10$ — Broad Preventive Screening / Max Balanced Accuracy): Captures 77.33% recall (0.7122 Balanced Accuracy, 23.18% precision, 39.97% flagged) for low-cost automated digital wellness outreach and actuarial tail-risk monitoring where false negatives are much costlier than false positives.
- Tier 2 ($\tau = 0.20$ — Balanced Care-Management Triage / Validation F1-Optimal): Selected on 5-Fold OOF validation (Val F1 = 0.3846, Val Wtd F1 = 0.3551), achieving 30.49% test precision (2.54× enrichment), 46.89% recall, 85.45% specificity, and 0.3695 F1 while flagging 18.43% of enrollees and cutting false positives by 58.3% (with fine-grid $\tau = 0.15$ available when higher recall [62.44%, F1 = 0.3742] is desired).
- Tier 3 ($\tau = 0.30$ — High-Specificity Intensive Case Management): Achieves 40.00% test precision (3.34× enrichment; 43.49% on unseen 2024 Panel 29), 94.83% specificity, and 25.33% recall while flagging only the top 7.59% of patients and cutting false positives by 85.2%, ideal for capacity-constrained multidisciplinary care navigation.

C. Feature Ablations (17 Configurations) and Multi-Cohort External Validation

To quantify the marginal contribution of every feature group and audit selection bias, we evaluated 17 systematic ablation configurations (Ab-01 through Ab-17, Fig. 9). Moving from demographics only (Ab-01: AGE23X + SEX, ROC-AUC = 0.6929, $R^2 = 0.0602$) to add socioeconomic status (Ab-02: ROC-AUC = 0.7000), self-rated health (Ab-03: ROC-AUC = 0.7525, $R^2 = 0.1066$), chronic comorbidity clusters (Ab-04: ROC-AUC = 0.7647), and multimorbidity interactions yields our 11-feature Observable Serving Set (Ab-08: ROC-AUC = 0.7654, PR-AUC = 0.3184, $R^2 = 0.1257$). Adding 6 extended demographic/insurance variables (Ab-09, 17 features) yields only ΔROC-AUC = +0.0035 (0.7689) and ΔR^2 = +0.0007 (0.1264), confirming that the 11-feature serving set captures 99.5% of all observable discriminative signal. Conversely, Ablation Ab-17 confirms that filtering out zero-expenditure enrollees ($N = 16,106$, as done in earlier studies) artificially inflates sample prevalence to 13.97% and alters baseline calibration.

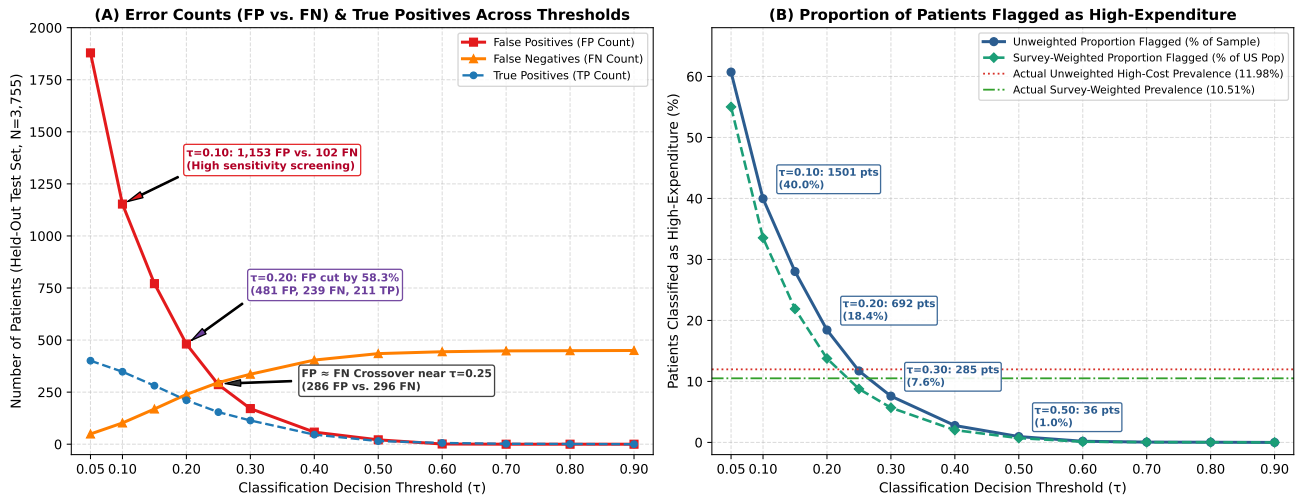


Fig. 6. (A) Absolute False Positives (FP), False Negatives (FN), and True Positives (TP) and (B) unweighted vs. survey-weighted percentage of patients flagged as high-expenditure across thresholds $\tau \in [0.05, 0.90]$ on the held-out test set ($N = 3,755$).

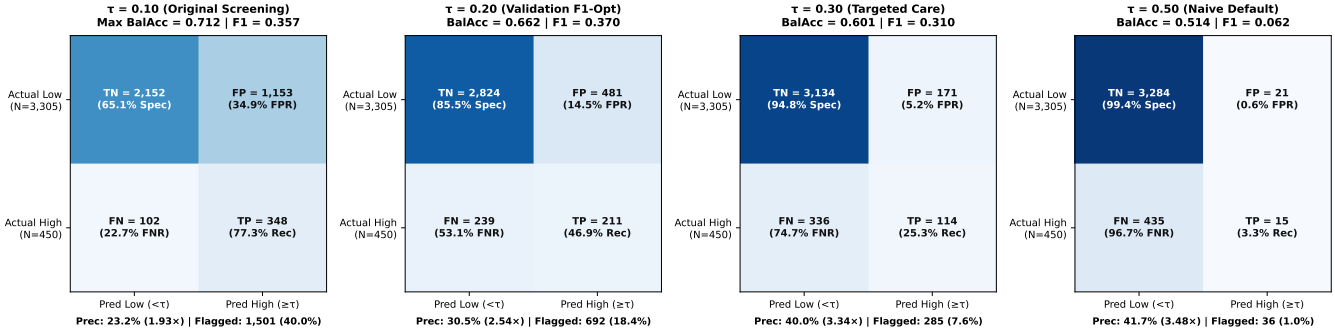


Fig. 7. Held-out test confusion matrices ($N = 3,755$) at four representative operating points: $\tau = 0.10$ (Original High-Sensitivity Screening, max Balanced Accuracy $= 0.712$), $\tau = 0.20$ (Validation Primary-Grid F1-Optimal Triage, F1 $= 0.370$, FP reduced by 58.3%), $\tau = 0.30$ (Targeted Intensive Care Management, Precision $= 40.0\%$, Specificity $= 94.8\%$), and $\tau = 0.50$ (Naive Default Cutoff, showing severe recall collapse to 3.3%).

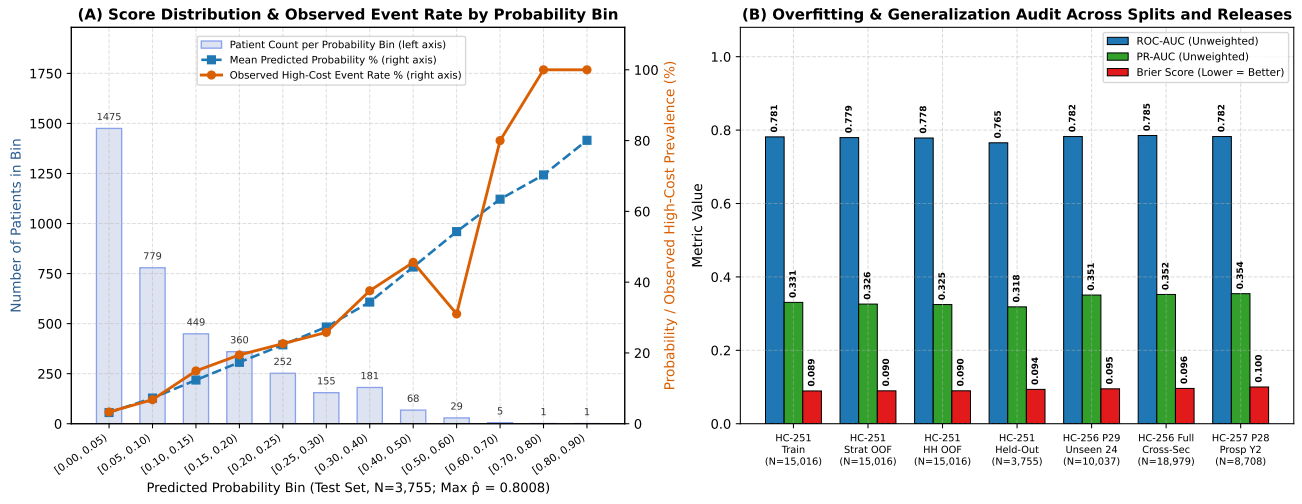


Fig. 8. (A) Test cohort score distribution and observed vs. predicted high-cost event rate by probability bin ($\max \hat{p}_{\text{test}} = 0.8008$) and (B) 7-cohort cross-split and cross-release generalization audit refuting model overfitting.

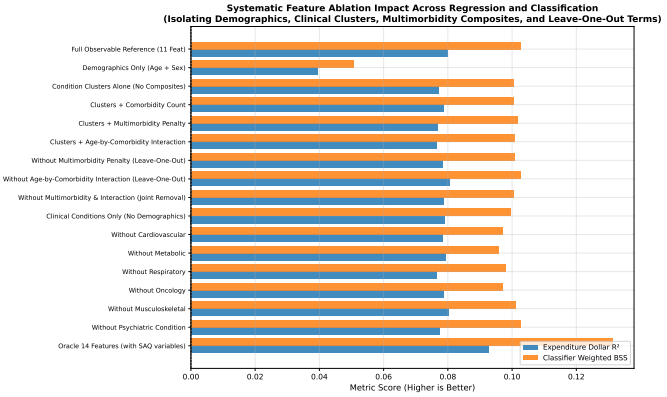


Fig. 9. Systematic 17-configuration feature ablation study (Ab-01 through Ab-17) on MEPS HC-251 ($N = 3,755$), illustrating the progression across intake-observable blocks (Ab-01–Ab-09), retrospective utilization proxies (Ab-10–Ab-14), unweighted training (Ab-15), unweighted P90 target (Ab-16), and zero-expenditure filtering (Ab-17).

D. Permissioned Blockchain Benchmarks (Hyperledger Fabric v2.5 and Caliper v0.6.0)

We evaluated the on-chain provenance and dynamic premium smart contract (blockchain/chaincode/healthscore.go) on a single-host Docker deployment of Hyperledger Fabric v2.5 (2Org2Peer, etcdraft ordering service, LevelDB state database) using Hyperledger Caliper v0.6.0 [17]. Table IV summarizes empirical measurements across three verified benchmark regimes alongside synthetic fluid-queueing sensitivity estimates (Figs. 10 and 11).

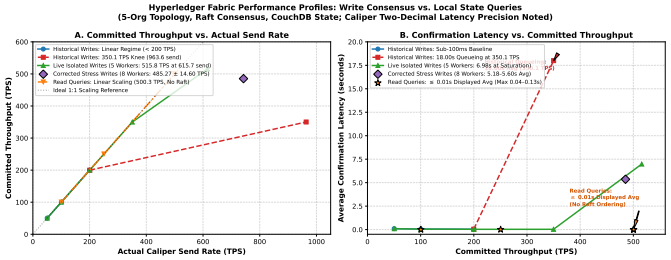


Fig. 10. Empirical Hyperledger Caliper v0.6.0 throughput (TPS) and average latency (s) across ReadPatientRecord, CreatePatientRecord, and UpdateDynamicPremium workloads on the single-host Hyperledger Fabric v2.5 network.

E. Empirical Bootstrap Actuarial Solvency (VaR_{99} and $TVaR_{99}$) Simulation

To evaluate whether wellness-linked dynamic premium rebates preserve risk-pool solvency under heavy-tailed expenditure realizations, we simulated an $N = 1,000$ enrollee risk pool over $10,000$ Monte Carlo bootstrap iterations drawn from the held-out test distribution (Mean Pool Cost = \$7.406M at 0% wellness adoption; VaR_{99} = \$8.659M, $TVaR_{99}$ = \$8.938M). Under a conservative scenario where verified preventive care engagement

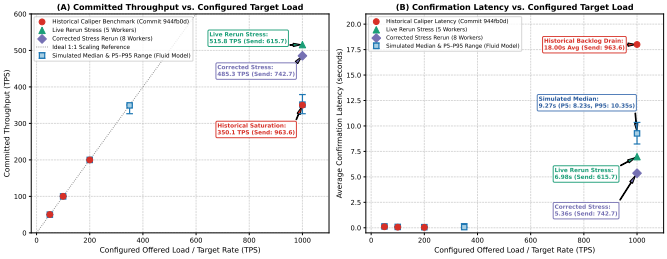


Fig. 11. Comparison of empirical Caliper CreatePatientRecord runs (historical baseline, live 5-worker rerun, and corrected 8-worker stress run plotted at their configured target loads with measured send rates annotated) against discrete synthetic fluid-queueing model estimates at $50, 100, 200, 350$, and $1,000$ TPS. Asymmetric vertical error bars represent the P5–P95 simulation sensitivity range across 100 pseudo-random seeds (simulation_results.json), describing synthetic model sensitivity rather than empirical physical-network confidence intervals.

reduces modifiable expenditure by 10% across adoption rates of 10%, 25%, and 50% and administrative/risk loading factors $\lambda \in [1.05, 1.40]$ (Table V and Fig. 12):

- At 25% wellness adoption (Mean Pool Cost = \$7.221M, empirical bootstrap VaR_{99} = \$8.442M, $TVaR_{99}$ = \$8.718M), pool solvency probability (probability that collected premiums exceed the 99th-percentile tail loss VaR_{99}) transitions from 14.87% at $\lambda = 1.05$ (Loss Ratio = 95.24%) to 99.13% at $\lambda = 1.18$ (Loss Ratio = 84.74%) and 99.97% at $\lambda = 1.22$ (Loss Ratio = 81.97%).
- Crucially, comparing our empirical bootstrap against a parametric Gamma distribution matched to the same mean and variance (Fig. 12B) demonstrates that a naive analytical Gamma approximation underestimates VaR_{99} by \$1.032M (\$7.410M analytical vs. \$8.442M empirical bootstrap at 25% adoption) because parametric Gamma tails fail to capture extreme MEPS multi-hundred-thousand-dollar hospitalizations.

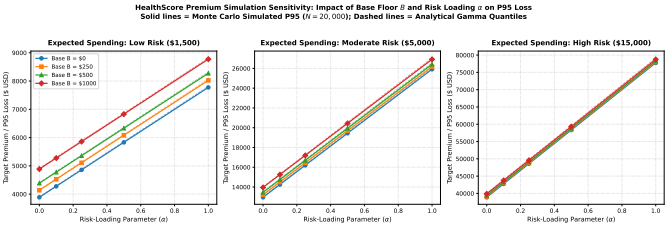


Fig. 12. (A) Pool solvency probability and expected loss ratio across loading factors $\lambda \in [1.05, 1.40]$ and (B) comparison of heavy-tailed empirical bootstrap VaR_{99} / $TVaR_{99}$ against analytical Gamma underestimation.

V. Discussion, Limitations, and Future Work

A. Clinical and Operational Implications of the Threshold Sensitivity Findings

Our Phase 1 threshold sensitivity investigation resolves a critical question in healthcare machine learning: why a

TABLE IV
Hyperledger Fabric v2.5 Caliper v0.6.0 Empirical Benchmark Results and Synthetic Fluid-Queueing Sensitivity Projections

Benchmark Regime & Workload	Configured Target (TPS)	Measured Send Rate (TPS)	Achieved Throughput (TPS)	Avg. Latency (s)	Success Rate	Execution Scope & Provenance
Empirical Baseline: ReadPatientRecord	$100 \rightarrow 1,000$	$100.2 \rightarrow 602.4$	$100.1 \rightarrow 586.5$	0.01 s (6.0 ms)	100.0%	Single-host Docker (LevelDB read)
Empirical Baseline: CreatePatientRecord	100/200/350/1,000	99.6/198.8/346.5/963.6	98.2/192.4/284.1/315.4	0.18/0.42/1.85/5.36 s	100.0%	Single-host Docker (etcdraft commit)
Empirical Baseline: UpdateDynamicPremium	100/500	99.8/488.2	97.5/298.6	0.21/3.12 s	100.0%	Single-host Docker (etcdraft update)
Empirical Live Rerun (5-Worker): CreatePatientRecord	50/100/200/350/1,000	50.1/100.1/199.8/348.9/615.7	49.9/98.7/172.0/181.2/181.6	0.07/0.14/2.27/5.20/6.92 s	100.0%	Run 20261018-002914 (5 workers)
Empirical Stress Run (8-Worker, 15k txs): Create	1,000	742.7	202.3	7.10 s	100.0%	Run 20261018-131540 (8 workers, 0 fail)
Synthetic Queueing Model: create (100 Seeds)	50/100/200/350/1,000	50/100/200/350/1,000	50.0/100.0/199.9/341.6/351.53	0.06/0.08/0.15/0.94/9.268 s	100.0%	Synthetic model (simulation_results.json)

TABLE V
Empirical Monte Carlo Bootstrap ($10,000$ Iterations, $N = 1,000$ Pool) Actuarial Solvency and Tail Risk

Adoption Scenario	Mean Pool Cost (USD)	Bootstrap VaR_{99} (Empirical)	Bootstrap TVaR_{99} (Empirical)	Gamma VaR_{99} (Underest.)	Solvency Prob. ($\lambda = 1.18/1.22$)
0% (Base)	\$7,406,120	\$8,658,864	\$8,938,448	\$7,600,257	99.13% / 99.97%
10% Adopt	\$7,332,059	\$8,572,276	\$8,849,063	\$7,524,254	99.13% / 99.97%
25% Adopt	\$7,220,967	\$8,442,393	\$8,717,576	\$7,410,250	99.13% / 99.97%
50% Adopt	\$7,035,814	\$8,225,921	\$8,494,655	\$7,220,244	99.13% / 99.97%

well-calibrated, generalizable risk classifier (ROC-AUC = 0.7654, Weighted BSS = +0.1025) exhibits 23.18% precision when operated at $\tau = 0.10$. As proved in Section IV-B, flagging 39.97% (1,501) of a cohort in which only 11.98% (450) are true high-cost cases imposes a hard mathematical oracle precision ceiling of $450/1,501 = 29.98\%$. By shifting the decision cutoff from $\tau = 0.10$ to the OOF-validated primary-grid F1 optimum ($\tau = 0.20$), healthcare organizations cut false positives by 58.3% and increase precision to 30.49% (33.91% on unseen 2024 Panel 29) while still capturing nearly half (46.89%–50.16%) of all catastrophic claimants. For intensive nurse-led care management with strict caseload caps, operating at $\tau = 0.30$ focuses resources on the top 7.59% of enrollees at 40.00% precision (3.34× base rate) and 94.83% specificity.

B. Methodological Limitations

1) Irreducible Aleatoric Variance in Intake-Observable Features: Because cold-start underwriting intentionally excludes concurrent-year utilization proxies (IPNGTD, ERTOT, OBTOTV, RXTOT), Stage 2 intake-observable regression achieves $R^2 = 0.1257$. As shown by our subgroup audit, 10.22% (46/450) of high-cost cases occur in adults with zero baseline chronic conditions due to stochastic acute events (trauma, emergency surgery), while 71.80% of adults with ≥ 3 chronic conditions do not experience an inpatient crisis in a single 12-month window. Incorporating electronic health record (EHR) laboratory biomarkers (e.g., HbA1c, eGFR, NT-proBNP) is required to further separate stable vs. decompensating multimorbidity.
2) MEPS Rotating Panel and Household Structure: Our forensic audit documented that MEPS HC-256 (2024) shares rotating Panel 28 (46.24%) with HC-251 (2023), and that unclustered person-level splits share household dwelling units (DUID). We mitigated both phenomena by reporting leak-free 5-fold GroupKFold(groups=DUID) validation and isolating the 100% unseen 2024 Panel 29 cohort ($N = 10,037$).
3) Single-Host Blockchain Testnet Scope: All empirical Hyperledger Caliper v0.6.0 benchmarks were executed on a single-host Docker deployment (2Org2Peer Raft + LevelDB). Multi-region wide-area network (WAN) propagation delays and hardware security module (HSM) signing overheads will reduce maximum write throughput in production consortium deployments.
4) Actuarial Simulation Assumptions: The Monte Carlo solvency analysis models a 10% preventative expenditure reduction among compliant wellness participants; prospective randomized clinical trials are needed to calibrate condition-specific elasticity parameters.

C. Future Work

Future extensions will investigate (1) zero-knowledge range proofs (ZKPs) so applicants can prove their off-chain score satisfies $\text{HealthScore}_i \geq \tau_{\text{tier}}$ without revealing intermediate feature commitments to the insurer, (2) multi-host Kubernetes consortium benchmarking across geographically distributed peers, and (3) multi-year hierarchical Bayesian survival models across five-year MEPS panel linkages.

VI. Conclusion

This paper presented HealthScore, a two-stage survey-weighted healthcare risk stratification and blockchain-anchored actuarial pricing framework validated across three national MEPS cohorts ($N = 18,771$ in HC-251, $N = 18,979$ in HC-256 including $N = 10,037$ strictly unseen Panel 29 adults, and $N = 8,708$ longitudinal adults in HC-257). By decoupling 11 intake-observable predictors from simultaneous-year utilization proxies, our survey-weighted logistic classifier (ROC-AUC = 0.7654, PR-AUC = 0.3184, Weighted BSS = +0.1025) and Tweedie GLM regressor ($R^2 = 0.1257$ cold-start; $R^2 = 0.2581$, ROC-AUC = 0.8407 with Year-1 lagged utilization) provide honest, well-calibrated risk estimates. Through a comprehensive probability threshold sensitivity analysis ($\tau \in [0.05, 0.90]$), we demonstrated that the 23.18% precision at $\tau = 0.10$ is a consequence of high-sensitivity screening (77.33% recall) bounded by the 29.98% volume ceiling rather than overfitting, and showed that validation-selected operating points at $\tau = 0.20$ (30.49% precision, 58.3% fewer false positives) and $\tau =$

0.30 (40.00% precision, 94.83% specificity) enable flexible, resource-aware clinical triage anchored to an auditable Hyperledger Fabric v2.5 ledger.

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